

Index laws



1 Write the following expressions in simplest index form:

a $2^{12} \times 2^9$

b $2^8 \times 11^8$

c $11^{12} \div 11^8$

d $21^5 \div 3^5$

e $(5^{12})^4$

f $15^{17} \div 15^8 \div 15^5$

g $(23^8)^9 \times 23^7$

h $\frac{(17^5)^8}{17^{32}}$

i $\frac{19^9 \times 19^4}{19^8}$

j $\frac{12^6}{12^4} \times 12^5$

k $\frac{(13^5)^2 \times 13^3}{13^5}$

l $\frac{(15^9)^5 \times 15^7}{15^{25}}$

2 Complete the following statements:

a $11^{11} \times 11^1 = 11^{19}$

b $5^{11} \times (1)^{11} = 35^{11}$

c $97^1 \div 97^{22} = 97^{12}$

d $55^6 \div (1)^6 = 11^6$

e $(11^4)^1 = 11^{12}$

f $7^8 \times 7^1 = 7^{14}$

g $2^{10} \times (1)^{10} = 10^{10}$

h $19^1 \div 19^{18} = 19^{20}$

i $60^4 \div (1)^4 = 12^4$

j $(13^8)^1 = 13^{16}$

3 Evaluate the following expressions:

a $6^5 \times 6^3$

b $7^3 \times 3^3$

c $4^8 \div 4^3$

d $(5^4)^2$

e $35^5 \div 5^5$

f $2^4 \times 4^4$

g $11^{18} \div 11^9 \div 11^7$

h $(3^3)^2$

i $\frac{6^5 \times 6^9}{6^{12}}$

j $7^{27} \div 7^{30} \div 7^3$

k $\frac{12^{10} \times 12^4}{12^{11}}$

l $\frac{(6^8)^6}{6^{46}}$

Negative bases

4 Write the following expressions in simplest index form:

a $(-11)^{10} \times (-11)^3$

b $(-7)^8 \times 3^8$

c $(-5)^2 \times 3^2$

d $(-3)^{12} \div (-3)^5$

e $(-12)^{20} \div (-12)^{19}$

f $(-30)^{50} \div (-30)^{47}$

g $(-48)^3 \div (-6)^3$

h $(-33)^{11} \div (-3)^{11}$

i $(-35)^5 \div 5^5$

j $(-42)^2 \div 7^2$

5 Complete the following statements:



a $11^3 \times (1)^3 = (-77)^3$

c $(-5)^1 \div (-5)^{31} = (-5)^{19}$

e $(-3)^1 \div (-3)^{39} = (-3)^{10}$

g $(-3)^7 \times (1)^7 = (6)^7$

i $(-33)^3 \div (1)^3 = -11^3$

b $(-5)^{11} \times (1)^{11} = 15^{11}$

d $(-14)^{13} \div (1)^{13} = (-7)^{13}$

f $(-5)^4 \times (1)^4 = (-60)^4$

h $(1)^3 \div (-3)^3 = (7)^3$

j $(1)^7 \div (-5)^7 = 11^7$

6 Evaluate the following expressions:

a $(-4)^{11} \div (-4)^7$

b $(-2)^3 \times (-2)^3$

c $(-3)^3 \times (-3)^2$

d $4^3 \times (-5)^3$

e $(-3)^8 \div (-3)^5$

f $15^5 \div (-3)^5$

g $2^3 \times (-3)^3$

h $(-14)^{11} \div 2^{11}$

i $(-7)^2 \times 5^2$

j $(-9)^4 \times (-3)^4$

k $(-100)^6 \div 50^6$

l $60^3 \div (-3)^3$

Fractional bases

7 Write the following in simplest index form:

a $\left(\frac{1}{3}\right)^4$

b $\left(\frac{3}{8}\right)^3$

c $\left(\frac{4}{16}\right)^8$

d $\left(\frac{15}{6}\right)^2$

e $\left(\frac{10}{33}\right)^5$

f $\left(\frac{2}{35}\right)^6$

g $\left(\frac{5}{18}\right)^3$

h $\left(\frac{29}{41}\right)^7$

i $\left(\frac{11}{13}\right)^9$

j $\left(\frac{20}{3}\right)^2$

k $\left(\frac{17}{4}\right)^4$

l $\left(\frac{31}{50}\right)^5$

8 Complete the following statements:

a $\frac{1}{27} = \left(\frac{1}{3}\right)^1$

b $\frac{64}{27} = \left(\frac{4}{3}\right)^1$

c $\frac{27}{8} = \left(\frac{3}{1}\right)^3$

d $\frac{1}{16} = \left(\frac{1}{4}\right)^2$

e $\frac{27}{8} = \left(\frac{3}{2}\right)^1$

f $\frac{1}{625} = \left(\frac{2}{5}\right)^4$

g $\frac{81}{100} = \left(\frac{1}{10}\right)^2$

h $\frac{256}{1} = \left(\frac{4}{5}\right)^4$

i $\frac{25}{9} = \left(\frac{5}{3}\right)^1$

j $\frac{1}{144} = \left(\frac{11}{12}\right)^2$

k $\frac{125}{27} = \left(\frac{1}{3}\right)^3$

l $\frac{32}{729} = \left(\frac{2}{3}\right)^1$

9 Evaluate the following expressions in fully simplified fraction:

a $\left(\frac{1}{3}\right)^2$

b $\left(\frac{3}{5}\right)^3$

c $\left(\frac{1}{2}\right)^5$

d $\left(\frac{7}{11}\right)^2$

e $\left(\frac{3}{8}\right)^3$

f $\left(\frac{12}{13}\right)^2$

g $\left(\frac{5}{9}\right)^3$

h $\left(\frac{2}{3}\right)^4$

i $\left(\frac{4}{5}\right)^6$

j $\left(\frac{2}{5}\right)^3$

k $\left(\frac{5}{7}\right)^2$

l $\left(\frac{10}{14}\right)^3$

10 State whether the following fractions are equal to $\left(\frac{1}{2}\right)^3$:

a $\frac{3}{6}$

b $\frac{1}{2^3}$

c $\frac{3}{2^3}$

d $\frac{1}{8}$

Negative indices

11 Is 2^{-3} less than or greater than 1?

12 Express the following with positive indices:

a a^{-9}

b $\frac{1}{a^{-n}}$

c $\frac{a^{-9}}{4}$

d $\frac{a^{-n}}{b^{-m}}$

e p^{-2}

f $3x^{-4}$

g $7x^{-9}$

h $p^{-2}q^3$

i $8p^{-3}$

j $2x^{-8}y^3$

k $\frac{b^{-7}}{6c^{-3}}$

l $m^{-5}n^{-4}p^4$

m $3a^4b^{-5}c^{-7}$

n $4^{-2}k^{-3}l^7$

o $\frac{f^{-7}}{g^{-2}h^8}$

p $\frac{s^4t^{-9}}{5u^{-6}}$

13 Express the following without fractions:

a $\frac{1}{u^4}$

b $\frac{2}{r^6}$

c $\frac{x}{y^2}$

d $\frac{4}{p^6q^7}$

14 Simplify the following, giving your answers with positive indices:

a $5y^9 \times 4y^{-3}$

b $7a^4 \times 4a^{-2}$

c $5x^4 \times (-3x^{-8})$

d $3y^{-2} \times 4y^{-3}$

e $2h^{-4} \times 4h^{11}$

f $3y^{-2} \times 2y^{-5}$

g $-4y^2 \times (-4y^{-5})$

h $(5mp)^2 \times mp^{-2}$

i $\frac{9x^2}{3x^9}$

j $\frac{15x^3}{5x^7}$

k $12x^5 \div 4x^{-3}$

l $9c^6 \div 6c^8$

m $\frac{15h^2}{12h^{-7}}$

n $\frac{12x^5}{4x^{-3}}$

o $\frac{21p^{-4}}{14p^{-3}}$

p $\frac{36w^{-6}}{16w^4}$

15 Simplify the following, giving your answer with positive indices:



a $(2m)^{-3}$

b $(4m^{-6})^4$

c $(3p^{-4})^{-2}$

d $(3y^2)^{-2}$

e $\left(\frac{y}{4}\right)^{-3}$

f $\left(\frac{x^5}{y^4}\right)^{-1}$

g $\left(\frac{x^7}{y^9}\right)^{-4}$

h $\frac{20x^3}{4x^{-2}}$

i $\frac{10x^{-7}}{2x^{-3}}$

j $\left(\frac{z}{3}\right)^{-4}$

k $\left(\frac{p^3}{q^7}\right)^{-1}$

l $\left(\frac{x^{-4}}{y^{-8}}\right)^{-2}$

16 Solve the following equations for n :

a $\frac{1}{25} = 5^n$

b $\frac{1}{8} = 2^n$

c $(x^3y^{-5})^n = x^{-12}y^{20}$

d $(a^{-5}b^3)^n = a^{15}b^{-9}$

17 Complete the following:

$$(x^4y^l)^l = \frac{y^{12}}{x^{16}}$$

18 A student was writing $5a^{-1}$ without negative indices and wrote $5a^{-1} = \frac{1}{5a}$. Explain why their working is incorrect, and write the correct answer.

19 Simplify:

a $2y^6 \times 4y^7 \times 4y^{-5}$

b $6y^7 \times 2y^{-5} \times 5y^3$

20 Simplify the following, giving your answers with positive indices:

a $\frac{2x^2y^{-4}}{7z^{-3}} \times \frac{5x^{-5}z^6}{3y^3}$

b $\left(\frac{3x^{-2}}{10y^4z^{-1}}\right)^{-2} \div \left(\frac{8x^{-5}z^2}{15y^2}\right)^{-1}$